Extra Practice 1: Absolute Value of Real Numbers

A. Skills

1. Evaluate.

(a) $|-31|$ (b) $-|-9.4|$ (c) $|12 - 5 \times 3|$ (d) $|-5| - |-12|$

(e) $6 - 2|-4|$ (f) $\big|\,|-10| - 15\,\big|$ (g) $-|\sqrt[3]{-216}|$ (h) $\sqrt{(-0.7)^2}$

2. Determine the distance between the numbers. Give exact answers.

(a) -12 and 5 (b) -8.5 and -1.25 (c) $\frac{2}{3}$ and $-\frac{3}{4}$ (d) $5\sqrt{2}$ and $-\sqrt{2}$

3. Arrange in order from greatest to least. $|-2.9|$, $-|-3|$, $|\sqrt{10} - 1|$, $\left|-\frac{8}{3}\right|$, $-|2 - 4.5|$

4. Evaluate each expression for $x = -2$ and $y = 3$.

(a) $|x - y|$ (b) $|x| - |y|$ (c) $|x^2 - 3y|$ (d) $\dfrac{|4x + y|}{|x|}$

B. Applying

5. A delivery robot moves along a straight sidewalk. Its positions (in metres, east of the start is positive) are 0, then 35, then -20, then 45, then back to 0. Determine its net change in position and the total distance it travels.

6. A runner aims for a lap time of 92 s. Her four lap times were 89.4 s, 93.1 s, 95.0 s and 90.8 s. Use $|t - 92|$ to find which lap was closest to the target and which was farthest from it.

7. Simplify.

(a) $\dfrac{x + |x|}{2}$ for $x > 0$, and for $x < 0$

(b) $|-4x|$ for $x < 0$

8. Which expression has the greatest value?

A. $|-3+8|$ **B.** $|-3|+|-8|$ **C.** $|-3|-|8|$ **D.** $-|-3-8|$

9. Numerical Response If $a=-4.5$ and $b=1.3$, the value of $|a-b|+|a|$ is _____.

C. Stretch

10. For which values of x is $|x-2|=x-2$? For which values is $|x-2|=2-x$?

11. Simplify $|x-3|+|x+1|$ without absolute value symbols when

(a) $x>3$

(b) $-1\leq x\leq 3$

(c) $x<-1$

12. The numbers a and b satisfy $a<0<b$, $|a|=2.5$ and $|a-b|=7$. Determine a and b.

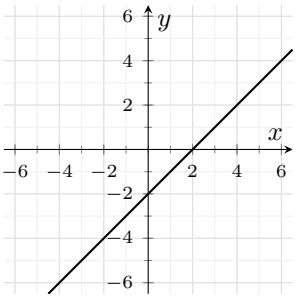
13. Is $|a+b|$ ever greater than $|a|+|b|$? Test pairs where both numbers are positive, both are negative, and the signs are opposite. Describe when $|a+b|=|a|+|b|$.

Extra Practice 2: Graphing Absolute Value Functions of Linear Functions

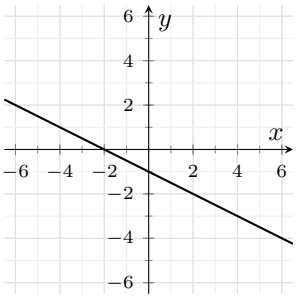
A. Skills

1. The graph of $y = f(x)$ is shown. Sketch $y = |f(x)|$ on the same grid and state the invariant point.

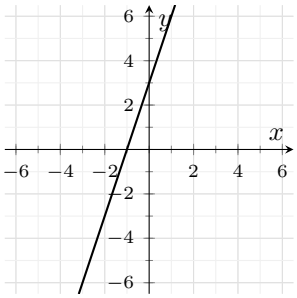
(a) $f(x) = x - 2$



(b) $f(x) = -\frac{1}{2}x - 1$



(c) $f(x) = 3x + 3$



2. For each function, state the vertex, the y-intercept, the domain and the range.

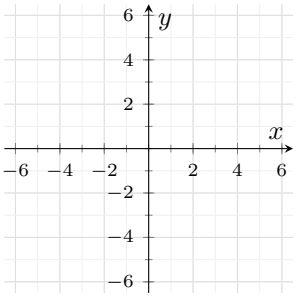
(a) $y = |x + 6|$ (b) $y = |5x - 15|$ (c) $y = \left|-\frac{3}{4}x + 3\right|$

3. Write each function in piecewise form.

(a) $y = |x - 9|$ (b) $y = |4x + 2|$ (c) $y = |12 - 3x|$ (d) $y = |-x - 1|$

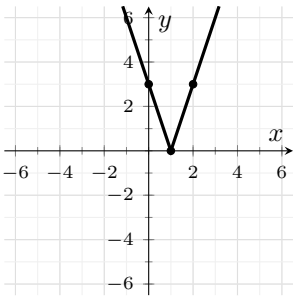
B. Applying

4. (a) Sketch $y = |2x + 3|$ on the grid. Label the vertex and the y-intercept.



(b) State the domain and range, and write the function in piecewise form.

5. The graph of $y = |f(x)|$, where f is linear, is shown. Write two possible equations for $y = f(x)$, and write $y = |f(x)|$ in piecewise form.



6. The function $y = f(x)$ is linear.

(a) The point $(-6, -2)$ is on $y = f(x)$. State the matching point on $y = |f(x)|$.

(b) The x-intercept of $y = f(x)$ is 2.5 and its y-intercept is -7. State the x- and y-intercepts of $y = |f(x)|$.

7. A car drives at a constant 100 km/h along a highway toward Red Deer, which is 150 km ahead, and continues past it. Its distance from Red Deer after t hours is $d = |150 - 100t|$ kilometres, for $0 \le t \le 2.5$.

(a) When does the car pass through Red Deer?

(b) How far from Red Deer is the car at $t = 2.5$?

(c) Sketch the graph (or use a calculator) to find when the car is 50 km from Red Deer.

(d) State the domain and the range of d in this context.

8. Which function has the same graph as $y = |2x - 8|$?

A. $y = |8 - 2x|$ **B.** $y = 2|x| - 8$ **C.** $y = |2x| - 8$ **D.** $y = -|2x - 8|$

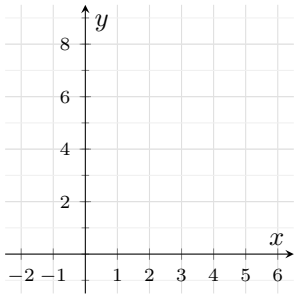
9. Numerical Response The graph of $y = |f(x)|$, where $f(x) = -5x + 13$, touches the x-axis at $(a, 0)$ and crosses the y-axis at $(0, b)$. The value of $b - a$, to the nearest tenth, is _____.

C. Stretch

10. Let $y = |x| + |x - 4|$.

(a) Write y without absolute value symbols for $x < 0$, for $0 \le x \le 4$ and for $x > 4$.

(b) Sketch the graph and state its range.



11. Use the graph of $y = |2x - 6|$ and a horizontal line $y = k$ to explain how many solutions the equation $|2x - 6| = k$ has when $k > 0$, when $k = 0$ and when $k < 0$.

12. Show that $|2x - 6| = 2|x - 3|$, and describe the graph of $y = |2x - 6|$ as a transformation of $y = |x - 3|$.

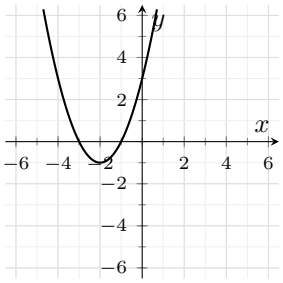
13. A linear function f has $f(-2) = 7$, and the graph of $y = |f(x)|$ has its vertex at $(1.5, 0)$. Determine $f(x)$ and the y-intercept of $y = |f(x)|$.

Extra Practice 3: Graphing Absolute Value Functions of Quadratic Functions

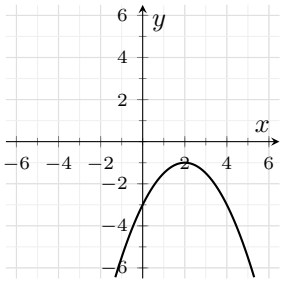
A. Skills

1. The graph of $y = f(x)$ is shown. Sketch the graph of $y = |f(x)|$ on the same grid and state the invariant points, if any.

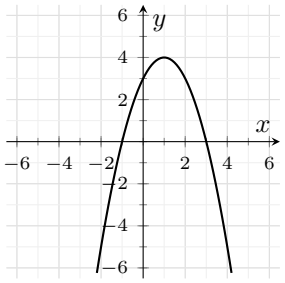
(a)



(b)

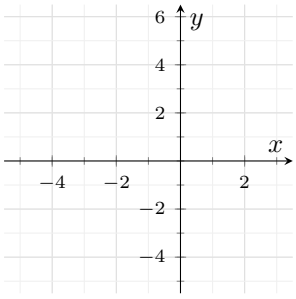


(c)

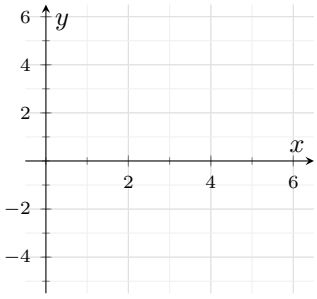


2. Sketch the graph of each function. Label the invariant points and the vertex of the graph.

(a) $y = |x^2 + 2x - 3|$



(b) $y = |-(x-3)^2 + 1|$



3. For each function, state the invariant points (if any), the y-intercept, the vertex of the graph and the range.

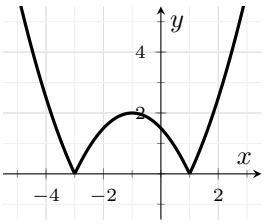
(a) $y = |x^2 - 7x + 10|$ (b) $y = |4(x+1)^2 + 3|$ (c) $y = |-(x-5)^2 - 2|$

4. Write each function in piecewise form, without absolute value symbols.

(a) $y = |x^2 + x - 20|$ (b) $y = |9x - x^2|$ (c) $y = |2x^2 + 3|$

B. Applying

5. The graph of $y = |f(x)|$ is shown, where f is a quadratic function. Write the two possible equations of f in factored form.



6. A quadratic function g opens up, has vertex $(2, -8)$ and passes through $(0, -6)$. Without graphing, state the invariant points, the y-intercept, the vertex of the middle arc and the range of $y = |g(x)|$.

7. A ball is thrown upward from the edge of a cliff. Its height relative to the cliff top is $h(t) = -5t^2 + 10t$ metres after t seconds, $0 \leq t \leq 3$ (negative values are below the cliff top). Its distance from the level of the cliff top is $d(t) = |h(t)|$.

(a) At what times is $d(t) = 0$?

(b) State the range of d for $0 \leq t \leq 3$.

(c) At what times is the ball 5 m from the level of the cliff top? Round to the nearest hundredth if needed.

8. Which function's graph has **no** invariant points?

A. $y = |x^2 - 4x + 4|$ **B.** $y = |x^2 - 4x + 3|$ **C.** $y = |-x^2 + 4x - 5|$ **D.** $y = |-x^2 + 4x|$

9. Numerical Response The y-coordinate of the vertex of the middle arc of the graph of $y = |2x^2 - 12x + 10|$ is _____.

C. Stretch

10. How many times does the line $y = k$ meet the graph of $y = |x^2 - 6x + 5|$? Answer for every value of k.

11. Can the graph of $y = |f(x)|$, where f is quadratic, have exactly one invariant point? If so, give two different examples of f and describe the graph of $y = |f(x)|$ for each.

12. f is a quadratic function that opens down. The graph of $y = |f(x)|$ has invariant points $(-2, 0)$ and $(6, 0)$ and y-intercept 6. Determine $f(x)$, and write $y = |f(x)|$ in piecewise form.

Extra Practice 4: Solving Absolute Value Equations (Linear)

A. Skills

1. Solve.

(a) $|x+8|=3$

(b) $|4x-2|=10$

(c) $5|x-1|+2=22$

(d) $|2x+7|+6=1$

(e) $-|x+4|=-9$

(f) $\left|\dfrac{x-2}{3}\right|=2$

2. Solve by cases.

(a) $|x-5|=4x$

(b) $|3x+6|=x+10$

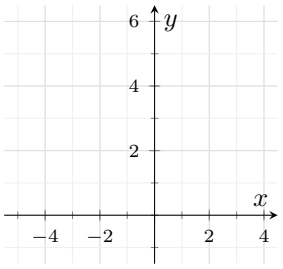
3. Solve and verify. Identify any extraneous solutions.

(a) $|x+3|=2x$

(b) $4|x-1|=2x+6$

(c) $|2x-1|+x=-2$

4. Solve $|x+1|=-x+3$ graphically. Sketch and label both graphs, then verify the solution algebraically. Explain why there is only one solution.



B. Applying

5. Describe how to solve $|2x+1|-3=x$ with the zero method on a graphing calculator, and give the solutions to the nearest hundredth. Then confirm them algebraically.

6. A ferry is scheduled to arrive at 14:30, which is 870 minutes after midnight. It is counted as on time if it arrives within 8 minutes of the schedule. Write and solve an absolute value equation for the earliest and latest on-time arrivals, and give them as clock times.

7. A hiker is at kilometre marker x on a straight trail that starts at marker 0 ($x \geq 0$). A ranger cabin is at marker 9. The hiker's distance from the cabin is half her distance from the start of the trail. Where could she be?

8. Solving $|x-2| = 3x+6$ by solving both linear equations gives $x = -4$ and $x = -1$. The solution of the absolute value equation is

A. $x = -4$ only **B.** $x = -1$ only **C.** $x = -4$ or $x = -1$ **D.** no solution

9. Numerical Response The sum of the solutions of $2|x+1| = x+8$, to the nearest hundredth, is _____.

C. Stretch

10. Solve $|x+1| = |2x-5|$. (Two numbers have the same absolute value when they are equal or opposite.) Why is no check needed for extraneous solutions here?

11. For which values of b does $|x| = x+b$ have no solution, exactly one solution, or infinitely many solutions? Use a sketch of $y = |x|$ and the line $y = x+b$.

12. Write an absolute value equation of the form $|2x+b| = c$ whose solutions are -1 and 4.

Extra Practice 5: Solving Absolute Value Equations (Quadratic)

A. Skills

1. Solve algebraically.

(a) $|x^2 - 12| = 4$

(b) $|x^2 + 2x| = 3$

(c) $|x^2 - x - 6| = 6$

(d) $|4x^2 - 9| = -1$

2. Solve $2|x^2 - 4x| - 5 = 3$. Give exact answers.

3. Solve $|2x + 5| = |x - 1|$.

B. Applying

4. Solve and check: (a) $|x^2 - 3x - 4| = x + 1$ (b) $|x + 3| = x^2 - 9$

5. Solve $|x^2 - 1| = 2x + 2$ and check.

6. Use a graphing calculator to solve $|x^2 - 5x + 2| = 3$ to the nearest tenth. Then give the exact solutions.

7. A company's monthly profit is $P = -2x^2 + 24x - 40$ thousand dollars when it makes x hundred units, $0 \leq x \leq 12$. Its target profit is \$20 thousand. For which production levels is the profit exactly \$10 thousand away from the target? Give exact answers and answers to the nearest unit.

8. How many solutions does $|x^2 - 2x - 3| = 4$ have?

A. 0 **B.** 2 **C.** 3 **D.** 4

9. Numerical Response The sum of all the solutions of $|x^2 - 7x + 6| = 6$ is _____.

C. Stretch

10. For which values of k does $|x^2 - 4| = k$ have (a) exactly four solutions, (b) exactly three, (c) exactly two? Explain with a sketch.

11. Solve $|x^2 - 4x| = |2x - 5|$. Give exact answers.

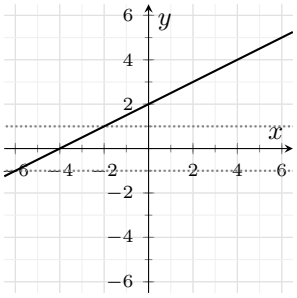
12. Solve $|x + 1| + |x - 3| = x^2$ using subdomains.

Extra Practice 6: Reciprocal Functions of Linear Functions

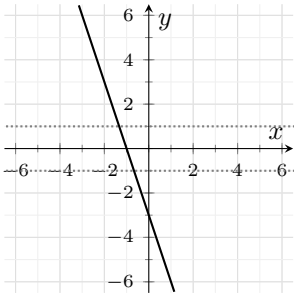
A. Skills

1. Sketch the graph of $y = \dfrac{1}{f(x)}$ on each grid. Label the vertical asymptote and the invariant points.

(a) $f(x) = \frac{1}{2}x + 2$



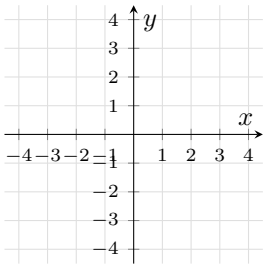
(b) $f(x) = -3x - 3$



2. For each function, state the vertical asymptote, the y-intercept, the invariant points, the domain and the range. (a) $y = \dfrac{1}{2x + 8}$ (b) $y = \dfrac{1}{3 - x}$

3. For $f(x) = 5x - 2$, evaluate $\dfrac{1}{f(x)}$ at $x = 0$, $x = 0.4$, $x = 1$ and $x = -2$.

4. Sketch $y = -x$ and $y = -\dfrac{1}{x}$ on the same grid (use a calculator to check). State the invariant points and the asymptotes of $y = -\dfrac{1}{x}$.



B. Applying

5. Complete the table for $y = \dfrac{1}{2x - 1}$ and describe the graph near $x = \frac{1}{2}$.

x	0.4	0.49	0.51	0.6
$\dfrac{1}{2x - 1}$				

6. The graph of $y = \dfrac{1}{f(x)}$, where f is linear, has the vertical asymptote $x = -2$ and the y-intercept $\frac{1}{4}$. Determine $f(x)$.

7. Describe how the graphs of $y = \dfrac{1}{x-1}$ and $y = \dfrac{1}{3x-3}$ are alike and how they are different.

8. Which statement about the graph of $y = \dfrac{1}{-2x+6}$ is true?

A. The vertical asymptote is $x = -3$.

B. The y-intercept is 6.

C. The graph is above the x-axis for $x < 3$.

D. For $x > 3$, y decreases as x increases.

9. Numerical Response For $f(x) = 3x - 4.5$, the graph of $y = \dfrac{1}{f(x)}$ passes through $(2, p)$. The value of p, to the nearest hundredth, is _____.

C. Stretch

10. The graph of $y = \dfrac{1}{f(x)}$, f linear, has the vertical asymptote $x = 4$ and the invariant point $(6, 1)$. Determine $f(x)$ and the other invariant point.

11. For $y = \dfrac{1}{x-k}$, the graph passes through $(5, 0.5)$. Determine k, and decide whether $(1, -0.5)$ is also on the graph.

12. Show that for any $f(x) = mx + b$, $m \neq 0$, the vertical asymptote of $y = \dfrac{1}{f(x)}$ is exactly halfway between the x-coordinates of the two invariant points.

Extra Practice 7: Reciprocal Functions of Quadratic Functions

A. Skills

1. State the number of vertical asymptotes of the graph of $y = \dfrac{1}{f(x)}$ and their equations.

(a) $f(x) = x^2 - 9x + 20$ (b) $f(x) = 9x^2 + 6x + 1$ (c) $f(x) = -x^2 - 3$ (d) $f(x) = 2x^2 + 7x - 4$

2. State the turning point of $y = \dfrac{1}{f(x)}$ and whether it is a maximum or a minimum.

(a) $f(x) = (x + 3)^2 + 4$ (b) $f(x) = -2(x - 1)^2 - 5$ (c) $f(x) = x^2 - 8x + 12$

3. (a) The point $(-1, 0.5)$ is on $y = f(x)$. State the matching point on $y = \dfrac{1}{f(x)}$.

(b) The point $(4, -1)$ is on $y = f(x)$. State the matching point on $y = \dfrac{1}{f(x)}$.

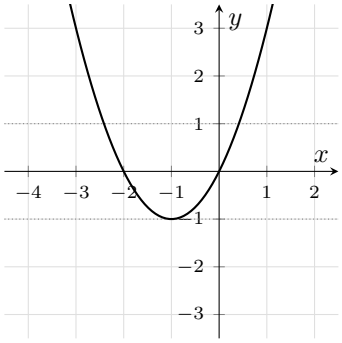
(c) The point $(2, 10)$ is on $y = \dfrac{1}{f(x)}$. State the matching point on $y = f(x)$.

4. State the domain and range of $y = \dfrac{1}{f(x)}$.

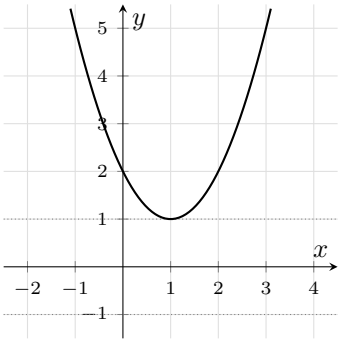
(a) $f(x) = (x - 2)^2 + 5$ (b) $f(x) = -3x^2$ (c) $f(x) = 9 - x^2$

5. The graph of $y = f(x)$ is shown. Sketch the graph of $y = \dfrac{1}{f(x)}$ on the same grid.

(a) $f(x) = x^2 + 2x$

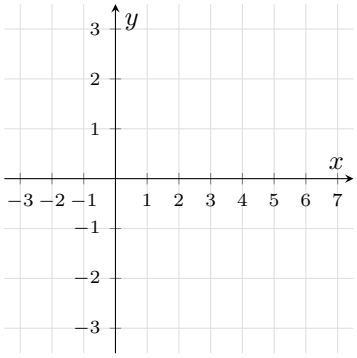


(b) $f(x) = (x-1)^2 + 1$



B. Applying

6. Let $f(x) = x^2 - 4x - 5$. For $y = \dfrac{1}{f(x)}$, determine the asymptotes, the y-intercept, the turning point and its type, the exact invariant points, and the domain and range. Then sketch the graph.



7. The graph of $y = \dfrac{1}{f(x)}$, where f is a quadratic function, has vertical asymptotes $x = 1$ and $x = 7$ and a minimum point $\left(4, \frac{1}{9}\right)$. Determine $f(x)$ in vertex form and in general form, and state the y-intercept of the reciprocal graph.

8. The graph of $y = \dfrac{1}{f(x)}$, where f is a quadratic function, has a maximum point $\left(-1, \frac{1}{4}\right)$ and passes through $\left(1, \frac{1}{12}\right)$. Determine $f(x)$ in general form.

9. For which function does the graph of $y = \dfrac{1}{f(x)}$ have no vertical asymptotes?

A. $x^2 - 4x + 4$ **B.** $x^2 - 4x + 3$ **C.** $x^2 - 4x + 5$ **D.** $-x^2 + 4x$

10. Numerical Response The graph of $y = \dfrac{1}{x^2 - 6x + k}$ has no vertical asymptotes and its maximum value is 0.25. The value of k is _____.

C. Stretch

11. For which values of k does the graph of $y = \dfrac{1}{x^2 + 2x + k}$ have (a) two vertical asymptotes, (b) one, (c) none?

12. The graph of $y = \dfrac{1}{f(x)}$, where $f(x) = ax^2 + bx + c$, has vertical asymptotes $x = -3$ and $x = 1$ and a y-intercept of $\frac{1}{6}$. Determine $f(x)$ and the turning point of the reciprocal graph.

13. Let $f(x) = x^2 - 2x - 3$. Describe how the graph of $y = \left|\dfrac{1}{f(x)}\right|$ is related to the graph of $y = \dfrac{1}{f(x)}$, and state its range.

Extra Practice 8: Unit 7 Review — Practice Test

Calculators are allowed. Give exact answers unless a question asks you to round. Verify the solutions of every absolute value equation.

A. Multiple Choice

1. The value of $|-4| - |-10| + |3|$ is

A. -3 **B.** 17 **C.** 3 **D.** -17

2. The distance between -7 and 3 on a number line is

A. 4 **B.** -10 **C.** 10 **D.** 21

3. Which statement is **false**?

A. $|x-2| = x-2$ if $x \geq 2$

B. $|1-3x| = 3x-1$ if $x > \frac{1}{3}$

C. $|x+4| = -x-4$ if $x < -4$

D. $|5-x| = 5-x$ if $x > 5$

4. The piecewise form of $y = |x^2 - 16|$ is

A. $y = \begin{cases} x^2-16, & \text{if } -4 \leq x \leq 4 \\ 16-x^2, & \text{if } x < -4 \text{ or } x > 4 \end{cases}$

B. $y = \begin{cases} x^2-16, & \text{if } x \leq -4 \text{ or } x \geq 4 \\ 16-x^2, & \text{if } -4 < x < 4 \end{cases}$

C. $y = \begin{cases} x^2-16, & \text{if } x \geq 4 \\ 16-x^2, & \text{if } x < 4 \end{cases}$

D. $y = \begin{cases} x^2-16, & \text{if } x \leq -4 \text{ or } x \geq 4 \\ x^2+16, & \text{if } -4 < x < 4 \end{cases}$

5. The range of $y = |x^2 + 2x - 3|$ is

A. $y \geq -4$ **B.** $y \geq 0$ **C.** $y \geq 4$ **D.** $0 \leq y \leq 4$

6. The complete solution of $8 - |x+1| = 3$ is

A. $x = 4$ **B.** $x = 4$ or $x = -6$ **C.** $x = -4$ or $x = 6$ **D.** no solution

7. The equation $|2x+3| = -1$ has

A. the solution $x = -2$

B. the solutions $x = -2$ and $x = -1$

C. the solution $x = -1$

D. no solution

8. The vertical asymptotes of the graph of $y = \dfrac{1}{x^2 - 9}$ are

A. $x = 3$ only **B.** $x = -3$ and $x = 3$ **C.** $x = 9$ **D.** $x = -9$ and $x = 9$

9. The graph of $y = \dfrac{1}{(x-2)^2+4}$ has

A. a maximum point at $\left(2, \frac{1}{4}\right)$

B. a minimum point at $\left(2, \frac{1}{4}\right)$

C. a maximum point at $(2, 4)$

D. a minimum point at $\left(-2, \frac{1}{4}\right)$

10. The point $(-2, 0.25)$ is on the graph of $y = \dfrac{1}{f(x)}$. Which point must be on the graph of $y = f(x)$?

A. $(-2, 4)$ **B.** $(-2, -4)$ **C.** $(2, 4)$ **D.** $(-2, -0.25)$

11. The points below are on the graph of $y = f(x)$. Which one must also be on both $y = |f(x)|$ and $y = \dfrac{1}{f(x)}$?

A. $(3, -1)$ **B.** $(2, 1)$ **C.** $(0, 0)$ **D.** $(-1, 2)$

B. Numerical Response

12. Numerical Response The product of the two positive solutions of $|x^2 - 7| = 2$, to the nearest hundredth, is _____.

13. Numerical Response The sum of the solutions of $|3x - 4| = x + 6$ is _____.

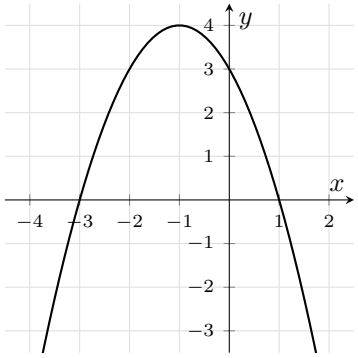
14. Numerical Response The graph of $y = \dfrac{1}{x^2 - 6x + k}$ has a maximum point $(3, -0.2)$. The value of k is _____.

15. Numerical Response Let $f(x) = x^2 - 2x - 7$. To the nearest hundredth, the smallest positive x-coordinate of an invariant point of the graphs of $y = f(x)$ and $y = \dfrac{1}{f(x)}$ is _____.

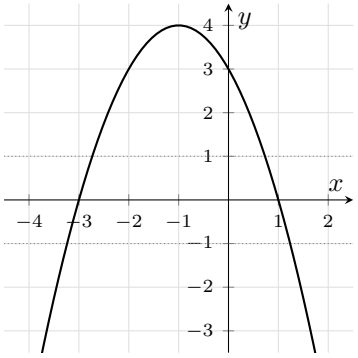
C. Written Response

16. The graph of $f(x) = -x^2 - 2x + 3$ is shown on both grids.

Grid 1: $y = |f(x)|$



Grid 2: $y = \frac{1}{f(x)}$



(a) Describe a strategy for graphing $y = |f(x)|$, and sketch it on Grid 1.

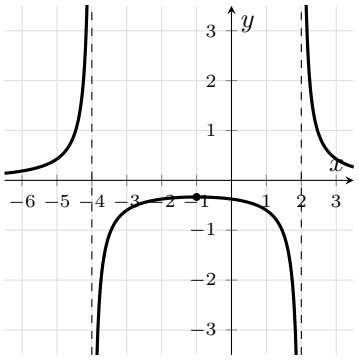
(b) Describe a strategy for graphing $y = \dfrac{1}{f(x)}$, and sketch it on Grid 2. Include the asymptotes, the turning point, the y-intercept and the exact invariant points.

(c) State the domain and range of $y = |f(x)|$ and of $y = \dfrac{1}{f(x)}$.

17. Solve algebraically. Identify any extraneous solutions.

(a) $|x - 6| = 2x$ (b) $|x^2 + x - 4| = 2$ (c) $|x^2 - 1| = x + 1$

18. The graph of $y = g(x)$, where $g(x) = \dfrac{1}{f(x)}$ and f is a quadratic function, is shown. It has vertical asymptotes $x = -4$ and $x = 2$ and a maximum point $\left(-1, -\frac{1}{3}\right)$.



(a) Sketch $y = f(x)$ on the grid.

(b) Determine $f(x)$ in the form $f(x) = a(x - p)^2 + q$.

(c) State the y-intercept of $y = g(x)$.

19. A mill cuts boards to a length of 2.4 m. A board is rejected if its length differs from 2.4 m by more than 0.015 m. Write an absolute value equation whose solutions are the shortest and the longest acceptable lengths, and solve it.

20. (a) Explain why the graph of $y = \dfrac{1}{f(x)}$ can never cross the x-axis.

(b) A student solved $|x + 2| = 2x - 5$ and gave the solutions $x = 7$ and $x = 1$. Explain the error.